



Oakwood University AI Studio & Engineering Lab

Turnkey proposal & execute-ready estimate for a 10-seat AI production studio, a hands-on engineering/soldering makerspace, and the department-wide power, network, camera, and access-control infrastructure that protects the investment — orchestrated end-to-end by Island Development Crew.

Prepared: June 2026

Client: Dr. Shushannah Smith
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What this is. A planning-grade, audit-aware General-Contractor proposal built from an on-site walkthrough (May 26, 2026), **71 verified site photographs**, and current June-2026 vendor pricing with named Huntsville subcontractors. IDC orchestrates every trade under one accountable point of contact, one transparent fee, and one progress dashboard. *Figures are planning ranges; each line is re-quoted to a sealed vendor/sub bid before purchase.*

10-Seat AI Studio · Rm 409

Soldering/Math Lab · Rm 407

Engineering · Rm 305 + 310

UniFi + pfSense Network

UniFi Protect Cameras

Multi-Door Access

≥4 Bambu Lab Printers

2084 / Lennox Curriculum

RECOMMENDED EXECUTION — ALL-IN (IDC-ORCHESTRATED)

\$172,000

Range \$170,625 – \$198,498. Includes equipment, subbed facility work, GC coordination, contingency, and the 10% IDC orchestration fee. Title III grant-eligible portion ≈ \$93.5K; facility/capital portion ≈ \$60.1K.

Mobilization deposit \$68,000 initiates procurement & sub scheduling (see Payment Schedule, p.8).

Secure payment: [Island Dev Crew Stripe payment link](#)
(\$68,000 mobilization deposit)

Signing: signature-ready PDF with direct blocks (p.8);
DocuSign / SignWell / Dropbox Sign accepted.

Department-Wide Vision (full robust build incl. dept cabling, dept cameras, Room-405 bathroom & raised floor) is on p.11 as the donor / naming-gift envelope at **\$245K–\$330K**.

The opportunity

Oakwood does not need another computer lab. It needs an **AI production studio** where Math & Computer Science students *ship* AI — apps, agents, fabricated hardware, and research artifacts — backed by reliable power, network, and security. As the only Seventh-day Adventist HBCU, Oakwood can also give students a credible **responsible-AI ethics** voice (curriculum, p.9).

Why IDC as GC

One orchestrator manages every vendor and Huntsville subcontractor (electrical, low-voltage, access, GC), holds a single **3–7% coordination markup** on subbed facility work and a transparent **10% orchestration fee**, and keeps faculty & administration informed through a live progress dashboard with weekly/monthly reports.

1 · Executive Summary & How to Read This

This package separates the project into the three quote families requested, because each draws on **different funding**. Selling all three as one grant line will be clawed back at audit.

Track	Scope	Planning range	Funding lane
Quote 1 — AI Studio + Engineering	Compute, Bambu fleet, soldering/CNC, AV + 3 boards, furniture, consumables, software	\$88.2K–\$98.8K	Mostly Title III grant-eligible
Quote 2 — Power / Electrical	FPE panel replacement, subpanel, dedicated circuits, raceway, raised floor, permits	\$54.4K–\$61.0K	Facility — NOT grant (capital / donor)
Quote 3 — Network + Cameras + Access	UniFi + pfSense, Protect cameras, 4-door Access; + structured cabling labor	Equip grant; cabling facility	Mixed

2 · Two Tiers — Recommended vs. Vision

Recommended Execution — ~\$172,000 all-in

Studio/lean compute, lab-focused facility scope (power + cabling to the labs + MDF; raised floor in the AI Studio only; *no* Room-405 bathroom). The number to commit to — top of the \$150–175K band you targeted; the execute-ready invoice (p.8) bills this.

Department-Wide Vision — \$245K–\$330K

Full robust transformation: pooled compute (RTX PRO 6000), dept-wide cabling (80 drops + fiber), dept-wide cameras, full electrical, raised floor *plus* the Room-405 bathroom and dept room-mod, 5-door access. The **donor / naming-gift** envelope (p.11).

3 · The Funding Firewall (critical)

Title III covers **supplies** (<\$2,000/unit) and **equipment** (≥\$2,000/unit) tied to the academic mission, with an audit trail and a **June 30** spend pressure. It does **not** cover **facilities** — electrical service, structural cabling labor, raised floor, construction, HVAC, the bathroom. Quote 2 in full and the cabling-labor / door-install portions of Quote 3 must come from **capital, physical-plant, donor, or match funds**.

4 · Sequencing — two hard dependencies + 90-day plan

- **No compute energizes before Quote 2 circuits.** The RTX workstation (575–600W) and DGX nodes need the dedicated 20A/30A circuits first.
- **The studio is unusable before Quote 3 drops.** Room 409 currently has *one* data drop; the 10GbE backbone + new drops come before deployment.

Phase	Actions
Now → June 30	Lock education-priced POs for Quote 1 (quotes expire 14–30 days); commission the licensed-electrician load study; campus-IT vendor ruling; confirm tax-exempt + line-by-line grant eligibility.
30 days	Sealed sub bids (3 each: electrical, low-voltage, access, GC); permits; cabling + raised-floor design; finalize room dimensions & seat count.
90 days	Install power → cabling/network → compute → cameras/access → demo day . Display boards + dashboard go live; first cohort onboards.

5 · Existing-Conditions Assessment (Walkthrough, May 26 2026)

A true on-site assessment was performed with Dr. Smith and is documented by all **71 verified photographs** (16 isolated as exhibits on pp.4–5; the full set with explanations is on the online resource page). Room → lab mapping:

Room	Becomes	Condition found (drives scope)
409	AI Studio (10–13 seats)	Preferred room. One data drop ; breakers trip on space heaters; keep dual smartboards (one Wacom) + 2 projectors; fire-egress concern; surface-run power on bare block.
407	Math & Soldering Lab	Convertible; blue steel pegboard workbenches + empty cabinetry + lab sink (reusable); an "L-bend" fits a fume extractor.
305 + 310	Engineering Lab	305 = project room (benches, existing MakerBot 3D printers). 310 = computer lab with rows of tower desktops (NVIDIA-GPU) to repurpose for AI hosting — saves the engineering budget.
MDF closet	Network core + power	Federal Pacific (FPE) panelboard "2C" with exposed bus + mixed 15/20/25/30A breakers, co-located with the network patch/fiber field (not to code); drops out ~monthly.

⚠ **Priority safety finding:** the MDF closet contains a **Federal Pacific Electric (FPE) panel** (EX-01-03) — a panel type with a documented industry history of breakers failing to trip on overload, a recognized fire hazard. Replacing it is strongly recommended and is carried in Quote 2. Whole-complex: no 90° angles, 15 "Frankensteined" AC units, windows with failed glazing/interpane fogging, bare cinderblock.

In Dr. Smith's words

"I've only got the one data drop right there." — (asked: in the entire room?) — "Yes, that's it right there."

"These electrical units, they can't handle nothing... I got portable heaters... Boom." → "That means your breakers need to be updated."

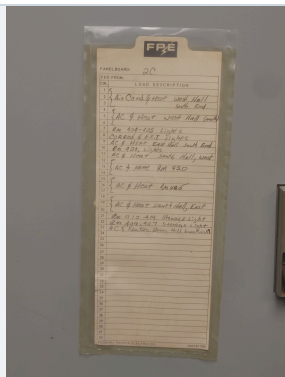
"I know this is not to code. We're not supposed to have a closet next to the fuse box."

"The main objective: secure and always changing... nobody could copy it or retain that information." (access control)

"We want AI. We need power. Power is the number one thing you invest in."

6 · Photo Exhibit Addendum — Part 1 of 2 (verified on-site)

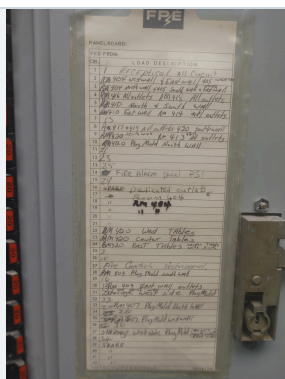
Each photograph was individually reviewed; captions describe what is actually in frame. **All 71 walkthrough photographs are included** at full resolution with explanations on the online resource page: [resources.html](#).



EX-01 · power · MDF/networking closet: FPE panelboard 2C directory card, handwritten, showing circuits dominated by AC/heat and lighting loads for the 400-series rooms and hallways (incl. Rm 404-405, 409, 420)



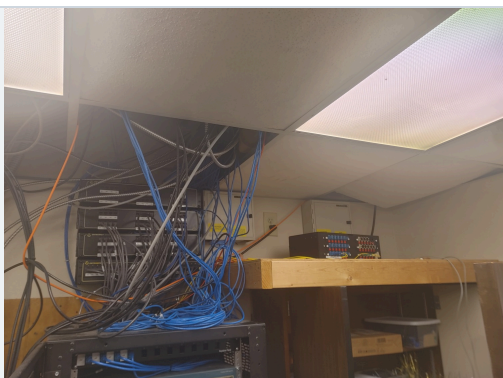
EX-02 · power · MDF/networking closet: Interior of FPE panel 2C with deadfront off: exposed bus bars and a mix of 15/20/25/30A single- and double-pole breakers hand-labeled to rooms 405, 409, 420 and hallway/A



EX-03 · power · MDF/networking closet: Sharp close-up of the FPE panel directory enumerating receptacle, plug-mold, dedicated-outlet, lab-table, fire-alarm, and '3D'-printer circuits across Rooms 404-420, with



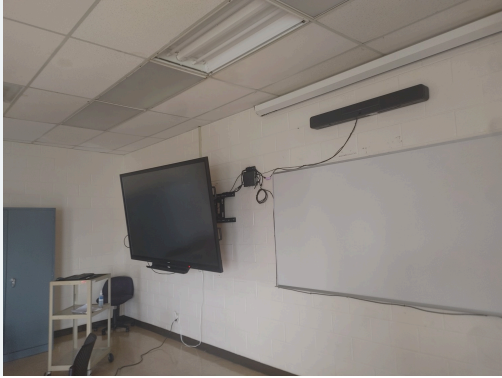
EX-04 · data-network · Hallway / classroom wall: Abandoned/exposed low-voltage drop: a surface-mount junction box with a loose twisted-pair (blue/yellow) hanging free and tucked behind the baseboard, not terminated in a



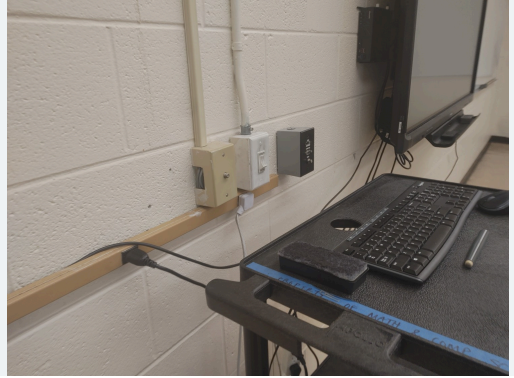
EX-05 · data-network · MDF/networking closet: MDF closet overview: a wall patch field and fiber breakout panel buried under unmanaged blue/gray/orange cabling routed through an open ceiling-tile gap, mounted on a bar



EX-06 · data-network · MDF/networking closet: Detail of the closet fiber distribution panel (rows A/B/C with red/blue couplers and yellow jumpers) beside a fiber-optic caution label and lone wall outlet, all on bare



EX-07 · AV · Classroom (AI Studio candidate): Primary AV wall: large interactive flat-panel display on an articulating mount, a soundbar above it, and a markerboard alongside, with exposed cabling and a wall-mounted



EX-08 · AV · 409: Mobile computer cart labeled for Math & Computer Science tethered by a single USB cable to a wall box, beside a wall-mounted flat panel with loose cabling. Shows reliance

7 · Photo Exhibit Addendum — Part 2 of 2



EX-09 · AV · Room 407 (Math & Soldering lab: Room 407 overview: rows of blue pegboard workbenches and stools beneath a recessed ceiling cassette AC unit and a ceiling-mounted projector, with two wall-mounted glass w



EX-10 · window · Classroom / exterior window: Window head/jamb corner showing chalky deposit and streaking at the glazing perimeter where the seal meets the frame. Documents the window-reseal need and moisture/efflor



EX-11 · window · Unspecified classroom/lab (dep: Window head detail showing failed glazing: pronounced interpane fogging across roughly half the pane and a marginal top seal against bare cinderblock, documenting the win



EX-12 · window · 409: Louvered wall grille over a window opening with bowed, misaligned blades and daylight leaking through gaps. Documents a damaged/ill-fitting louver assembly and a fenestra



EX-13 · bench · 407: Room 407 lab casework: continuous black-topped bench with stainless sink, faucet, and wood cabinetry. Only a single surface-mounted countertop duplex receptacle serves th



EX-14 · bench · Room 407 (Math & Soldering lab: Detail of Room 407 workbench assets: heavy blue steel benches with pegboard backboards, lower storage shelves, and matching wheeled stools. Clean documentation of reusabl



EX-15 · printer · Rooms 305/310 (Engineering): Crisp close-up of the engineering 3D printers, clearly identifying the MakerBot Replicator and a second enclosed unit. Best documentation of the existing 3D-printer asset



EX-16 · printer · Engineering computer lab (Room): Wide view of a fully equipped computer classroom: rows of tower desktops with dual flat-panel monitors on shared desks, blue task chairs throughout. Strong asset exhibit

These 16 are the strongest exhibits; the complete set of all 71 walkthrough photographs — with room/condition explanations — is on the online [resource & references page](#).

8 · Quote 1 — AI Studio + Engineering Lab (Title III grant-eligible)

AI Studio · Rm 409 10–13 high-memory seats so students run local models: 6× Framework Desktop 128GB (~\$2,851) + 4× Mac Studio M4 Max 64GB (~\$2,600) + 1× DGX Spark (\$4,699) + an RTX 5090 workstation + Jetson AGX Thor + Orin Nano kits + Synology NAS + rack UPS + an 85" demo wall. Keep the dual smartboards/2 projectors.	Math & Soldering · Rm 407 5–10× Hakko FX-888DX soldering stations + recirculating BOFA 3D PrintPRO-class fume/VOC extraction in the cabinetry/L-bend; bench instruments; an enclosed Carbide Nomad 3 CNC. Reuses the existing blue pegboard benches + cabinetry.	Engineering · Rm 305 + 310 ≥4 Bambu Lab printers: H2D large-format + X1E enterprise + 3× P1S + AMS, enclosures, filament. Repurpose the existing NVIDIA-GPU tower desktops in 310 for AI hosting (10GbE NICs, RAM, IP-KVM) — engineering compute for near-zero new spend.
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Division	Representative real SKUs (June 2026)	Planning range
AI Studio compute (lean)	Framework Desktop 128GB · Mac Studio M4 Max 64GB · DGX Spark · RTX 5090 WS · Jetson Thor/Orin · Synology DS1825+ · rack UPS	\$30,000–\$33,000
Engineering Bambu fleet + GPU repurposing	Bambu H2D AMS Combo ~\$2,199 · X1E ~\$2,649 · 3× P1S+AMS ~\$799 · BOFA PrintPRO 3 · Intel X550-T2 NICs · PiKVM V4	\$16,900–\$18,500
Soldering bench + CNC	5–10× Hakko FX-888DX (~\$135) · bench fume extraction · Carbide Nomad 3 ~\$2,800	\$8,000–\$9,000
AV: 3 hallway display boards + 85" studio demo wall	3× ~50" commercial displays + UniFi Connect signage/media players + mounts · 85" demo wall ~\$4,600	\$8,250–\$9,120
Furniture (movable) · Consumables/safety · Software/warranty	Movable benches/chairs · filament/resin/PPE/flammables cabinet/ABC+Class-D extinguishers · licenses + extended warranty	\$9,800–\$11,900
Quote 1 grant-eligible subtotal (equipment + supplies)		\$88,242–\$98,820

Display / project boards (your spec): a board outside *each* of the three labs (409 / 407 / 305-310), auto-fed from in-room activity via UniFi Connect — "you see AI happening before you enter the room." The **85" demo wall** + per-semester access make the studio look intentional and impressive. **Grant tag:** items ≥\$2,000/unit are EQUIPMENT; the many <\$2,000 items are SUPPLIES — confirm with the grants officer whether rack components aggregate to a capitalized system.

9 · Quote 2 — Power / Electrical (Facility · NOT grant · subbed + GC-managed)

Scope (lab-focused)	Planning range
Replace the FPE panel · 100A lab subpanel · 14 dedicated 20A circuits · 2× 30A L5-30 UPS circuits · conduit/raceway on cinderblock · lab touch-ups · permits	\$34,000–\$37,000

Load calc (electrician verifies): ~7–9kW continuous / ~12kW peak (~60–75A @120V) → a dedicated 100A subpanel is the floor (NEC 125% continuous derating; resolve 120/240V vs 208V; size the CNC circuit for inrush). **The Federal Pacific panel (EX-01-03) should be replaced regardless of the AI build** — a standalone life-safety item. **Named Huntsville subs to bid:** Madison Electric, Pro Electric, Lee Company, Blake Brothers, Urban Electric.

10 · Quote 3 — Network + Wi-Fi + Cameras + Access (Mixed: electronics grant / cabling facility)

Division	Real SKUs (June 2026)	Range	Lane
UniFi core + pfSense	UniFi EnterpriseXG-24 (\$1,299) · 2× USW-Pro-Max-24-PoE (\$799) · UDM-Pro-Max (\$599) · Netgate 6100 MAX pfSense (\$999) · 42U rack	\$3,837–\$4,500	Grant
UniFi Protect cameras	UNVR-Pro (\$499) + 8TB HDDs · 10–12× G5/G6 Turret/Bullet (\$129–\$199) · AI Pro (\$499) for the print room	\$3,055–\$3,800	Grant
UniFi Access — 4 doors, rotating credentials	4–5× UA-Hub-Door (\$199) · G3 Reader Pro (\$379) + G3 Readers (\$159) · fail-safe HES strikes/maglock + REX + DPS + install	\$8,400–\$9,000	Equip+install
Structured cabling — lab-focused (SUBBED)	Cat6A drops + patch panel + MDF rack rebuild (cabling labor = facility)	\$12,000–\$14,000	Facility

Per-semester rotating credentials are native to UniFi Access (PIN expiry, NFC deactivation, mobile/Wallet revoke). **Named Huntsville subs:** low-voltage — Travel Tech (256.970.4558), Interweave (256-837-2300), Heart Technologies (256-859-4000), State Systems; access — AAC Systems, Clear Link, LOCKTEC (256-881-9300), Tri-State Electrical. **ADA/egress:** all locks fail-safe; maglocks tie to the fire panel with REX + push-to-exit; Huntsville AHJ sign-off; low-voltage permit carried in Permits.

11 · Execute-Ready Invoice — Recommended Execution

Invoice basis: IDC (Jonathan Isaac) as General Contractor. Grant-eligible lines may be procured under Title III; facility lines are subbed and carry GC coordination. Re-quote each line to a sealed bid before purchase.

#	Line item	Amount (range)
I · Grant-Eligible Equipment & Supplies (Title III)		
1	AI Studio compute — studio/lean tier	\$30,000–\$33,000
2	Engineering Bambu fleet + GPU-PC repurposing	\$16,900–\$18,500
3	Soldering bench + CNC	\$8,000–\$9,000
4	Consumables / safety / storage	\$3,000–\$3,800
5	AV — 3 hallway display boards + 85" studio demo wall	\$8,250–\$9,120
6	UniFi network core + pfSense edge	\$3,837–\$4,500
7	UniFi Protect cameras — priority labs + key hallway	\$3,055–\$3,800
8	UniFi Access — 4 doors, per-semester rotating credentials	\$8,400–\$9,000
9	Furniture (movable lab/studio)	\$5,000–\$5,800
10	Software / licensing / extended warranty	\$1,800–\$2,300
Grant-eligible subtotal		\$88,242–\$98,820
II · Facility — Capital / Donor (NOT grant-eligible · SUBBED)		
11	Power / electrical — FPE replacement, subpanel, circuits, raceway, permits	\$34,000–\$37,000
12	Structured cabling — lab-focused (Cat6A drops, patch, MDF rebuild)	\$12,000–\$14,000
13	Raised access floor — AI Studio only (~300–400 sqft) + light room mod	\$8,000–\$10,000
14	Permits / insurance / freight	\$2,400–\$2,800
Facility subtotal		\$56,400–\$63,800
Direct cost (I + II)		\$144,642–\$162,620
III · GC Coordination, Contingency & IDC Fee		
15	GC coordination markup — 3–7% on the subbed facility base only	\$1,692–\$4,466
16	Contingency — 6–8% on (direct + markup); 2026 memory-price volatility	\$8,780–\$13,367
17	IDC Orchestration Fee — 10% of pre-fee subtotal	\$15,511–\$18,045
ALL-IN TOTAL · Recommended commitment \$172,000		\$170,625–\$198,498

Payment schedule (proposed)

- **\$68,000 mobilization** — on signature; initiates procurement & sub scheduling.
- **~50% progress** — at equipment delivery + electrical/cabling rough-in.
- **Balance at completion** — commissioning, demo day, & closeout docs.

Pay deposit: [IDC Stripe payment link](#) (\$68,000)

Acceptance & signatures

Signature-ready; DocuSign / SignWell / Adobe / print-scan accepted.

Provider

Jonathan Isaac
Island Development Crew LLC
jon-isaac@islanddevcrew.com

Signature / Date

Client

Dr. Shushannah Smith
Oakwood University · Math & CS
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Signature / Date

12 · Curriculum Spine — 2084 / John Lennox (responsible-AI ethics for an Adventist HBCU)

Textbook: "**2084 and the AI Revolution**" (Updated & Expanded Ed., John C. Lennox, Zondervan Reflective 2024). The class meets **twice weekly** (e.g., Mon/Wed or Tue/Thu); **one chapter is discussed per session — ~2 chapters/week**, so the cohort reads *through the whole book* over the term while every 1–2 weeks ships one of **8 portfolio products**. Two interleaved courses (**AI Engineering + hands-on Machining/CNC**) share this one ethics seminar. Pricing & outcomes on p.10.

Wk	Ethics seminar — Lennox (≈2 ch/wk)	AI Engineering room (Rm 409 / 310) + fabrication (407 / 305)	Product shipped
1	Ch 1–2 · Mapping AI + where we come from	Onboarding · Python/Git · first <i>local</i> LLM on the Rm 310 GPUs; local-vs-cloud	— setup
2	Ch 3–4 · Trajectory / surveillance + narrow-AI promise	Build a RAG assistant on a local model; prompt & retrieval basics	P1 Local RAG assistant
3	Ch 5–6 · Narrow-AI harms & bias + upgrading humans	CV classifier + bias/fairness audit; model cards	P2 CV model + bias audit
4	Ch 7–8 · AGI risk + what is a human being	Agentic workflow (tool-use) + guardrails; prompt-injection red-team	—
5	Ch 9–10 · Origin of moral sense + the true Homo Deus	Voice/assistive-tech device — AI + a 3D-printed part on the Bambu fleet	P3 Assistive-tech AI device
6	Ch 11–12 · Future shock / revelation	Multimodal app; deploy to Vercel with a Supabase backend	P4 Deployed multimodal app
7	Ch 13 + synthesis · responsible AI & the time of the end	Eval harness + red-team; responsible-AI governance memo; demo-board publish	P5 Eval + governance memo
8	Applied sprint · ethics review embedded in code review	Business / SaaS build: auth, database, billing	P6 SaaS product
9	Applied sprint	Agentic product via OpenRouter multi-model routing (local + frontier)	—
10	Publication / research artifact	Technical write-up + research poster; hallway demo board	P7 Publication / poster
11	Capstone build	Ship a real campus tool (à la the OU Campus Companion)	—
12	Capstone demo day	Polish + portfolio site; present the 8-product portfolio	P8 Capstone + portfolio

Reading completes the core 2084 arc by ~week 7; weeks 8–12 are build sprints with the ethics lens carried into every code review. The fabrication track (soldering / 3D-print / CNC) runs in parallel so several of the 8 products are physical-AI devices, not just software.

13 · Curriculum — Per-Student Toolchain, Pricing & Outcomes

Per-student digital toolchain (~\$200–\$250 / semester)

Tool	/ mo	What it unlocks
ChatGPT Plus	\$20	OpenAI frontier models
Claude Pro	\$20	Anthropic frontier models
Google Gemini	\$20	Google frontier models
xAI Grok	\$20	xAI frontier models
OpenRouter credit	\$20	One API → local, Chinese & all frontier models
Vercel	~\$0–20	Ship/host student apps
Supabase	~\$0–25	Auth + database backend
≈ \$100/mo subscriptions + hosting		= \$200–\$250 / student for the semester

This gives every student the **top four frontier labs + local & open models** via one toolchain — enough to design, build, deploy, and host real products.

All-in student price & the outcome

- **All-in: \$300–\$350 / student** — the **\$200–\$250 digital toolchain** + course materials + the *2084* book (combined; no separate book charge).
- **The deliverable:** by the time they leave the class — and certainly by graduation — **each student has 8 fully functioning products to their name:** an app, a business, software, a physical-AI device, or a publication.
- **The promise:** students graduate *job-ready and able to ship* — a sharpened, market-relevant portfolio instead of just a résumé.

Why it lands at Oakwood

As the only SDA HBCU, Oakwood can teach AI with a **worldview-grounded ethics voice** secular schools struggle to articulate — data dignity, human-in-the-loop, Sabbath as a model for digital rest. Graduates leave with a portfolio *and* a point of view.

14 · Fundraising, Match Funds & the Naming Opportunity

Grant & corporate funding • Federal HBCU + AI: NSF ExpandAI, NSF HBCU-EiR, NSF HBCU-UP, Dept of Ed Title III Part B (Oakwood is an active recipient). • Compute/hardware programs: NVIDIA Academic Grant, AMD University Program, Google.org / UNCF. • Mission-fit philanthropy: John Templeton Foundation, Lilly Endowment (faith & AI), Mellon ("Unruly Intelligences"). • Corporate matching-gift programs.	Notoriety & the naming track • Voices to engage: Dr. James Tour, John Lennox, Wesley Huff — credible on AI + faith + science. • Channels: Modern Wisdom, Valuetainment, Diary of a CEO — to spotlight an HBCU building responsible AI. • Naming opportunity: "The John Lennox AI Innovation Lab" (pending Oakwood approval) — pairs the 2084 curriculum with a named-gift envelope that funds the grant-ineligible facility scope.
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15 · Department-Wide Vision (donor / naming ask · \$245K–\$330K)

Division (full scope)	Range	Division (full scope)	Range
AI Studio compute (pooled · RTX PRO 6000)	\$42K–\$50K	Power / electrical (full dept)	\$34K–\$48K
Engineering Bambu fleet (full)	\$20K–\$23.9K	Structured cabling (80 drops + fiber)	\$24K–\$36K
Soldering + CNC (full)	\$11K–\$14K	Raised floor + Room-405 bathroom + room mod	\$45.6K–\$62K
Cameras (dept-wide) + Access (5 doors)	\$15K–\$19K	Network core, AV, furniture, software, permits	\$24K–\$30K
Department-Wide Vision — all-in (incl. GC markup + 10% IDC fee)			\$245,000–\$330,000

16 · Primary Source Links & References

Every quoted price traces to a named vendor or published benchmark (current June 2026). Re-quote each line to a sealed bid before purchase; 2026 memory-market pricing is time-sensitive.

AI / Compute • Oakwood AI Lab — final web reference • NVIDIA DGX Spark — marketplace • NVIDIA DGX Spark — product • Framework Desktop • Apple Education Institution Price List • NVIDIA Jetson Orin Nano Super • NVIDIA Jetson AGX Thor • Synology DiskStation	Engineering / Fabrication • Bambu Lab P1S • Bambu Lab H2D (large format) • Bambu Lab X1E (enterprise) • Original Prusa XL • Hakko FX-888DX soldering station • Hakko fume extraction • Donaldson BOFA 3D PrintPRO • Carbide 3D Nomad 3 CNC	Network / Security • Ubiquiti UniFi store • UniFi U7 Pro Max AP • UniFi EnterpriseXG-24 (10GbE core) • UniFi Dream Machine Pro Max • Netgate 6100 MAX (pfSense+) • UniFi Protect cameras • UniFi Access (door control)
Student toolchain / curriculum • ChatGPT (OpenAI) • Claude (Anthropic) • Google Gemini • xAI Grok • OpenRouter (one API, all models) • Vercel (hosting) • Supabase (backend/DB) • 2084 and the AI Revolution (Lennox, 2024)	Facility benchmarks • Ethernet / structured cabling install cost • Dedicated circuit install cost • Raised access floor cost guide	

Named Huntsville subcontractors (for the facility scope)

Trade	Named Huntsville-area subcontractors (solicit 3 sealed bids each)
Electrical	Madison Electric Inc · Pro Electric Inc · Lee Company · Blake Brothers · Urban Electric LLC
Low-voltage / cabling	Travel Tech (256.970.4558) · Interweave Technologies (256-837-2300) · Heart Technologies (256-859-4000) · State Systems Inc.
Access control / locks	AAC Systems LLC · Clear Link Systems · LOCKTEC Inc. (256-881-9300) · Tri-State Electrical Contractors
General contracting / raised floor	Regional Builders (256-684-8833) · Wilcock Builders · Lee Builders · Pro Access Floors · Southwest Solutions Group · Intertech Commercial Flooring

Full audited bill of materials with per-line SKUs and source URLs is in the package price book; all 71 walkthrough photographs at full resolution with explanations are on the online [resource page](#).

17 · Orchestration, Reporting & Next Steps

Single point of accountability IDC engages and schedules every vendor & Huntsville sub, manages the grant vs. facility funding lanes, and owns the timeline. One markup, one fee, one invoice.	Live progress & reporting A live HTML progress dashboard link is attached to the project email; weekly & monthly progress reports keep faculty & administration abreast through commissioning.	Protected investment Department cameras + per-semester rotating access control + cabling/power done to code (incl. FPE panel replacement) mean the equipment is controlled, managed, and protected from day one.
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Open items to confirm before sealed bids

- Final room dimensions & seat count (Rm 409 measured pass) — required for the raised-floor quote.
- Campus-IT vendor ruling: UniFi/pfSense allowed, or map to Aruba/Cisco/Extreme? Does access tie to the campus card?
- Oakwood tax-exempt status + grant eligibility ruling (setup labor, software/SaaS, networking hardware, cabling labor, access hardware).
- Tier to commit: Recommended ~\$172K vs. a sub-\$170K trim (single-lab cameras / 2-door access).

Recommended next step. Approve the Recommended Execution tier (~\$172,000 all-in), release the \$68,000 mobilization deposit (Stripe link, p.1 / p.8), and authorize IDC to (1) commission the licensed-electrician load study and (2) issue three sealed sub bids per trade. IDC returns firm, signed numbers within ~10 business days and begins the 90-day build.

Online: [Package hub](#) · [All 71 photos + references](#) · [Live dashboard](#) · [Pay \\$68K deposit](#)

Authorized to proceed — Provider Jonathan Isaac · Island Development Crew LLC _____ Signature / Date	Authorized to proceed — Client Dr. Shushannah Smith · Oakwood University _____ Signature / Date
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